

2201NSC Linear Algebra and Applications

Week 10

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Affine Transformations

Translations, homogeneous coordinates and composition

Affine transformations combine a linear map and a translation

An affine transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ has the form

$$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b},$$

where $A \in \mathbb{R}^{n \times n}$ and $\mathbf{b} \in \mathbb{R}^n$.

Linear part: $A\mathbf{x}$

Scaling, rotation, reflection and shear can all be represented by a matrix.

Translation part: $\mathbf{b}$

Every point moves by the same displacement. In particular, $T(\mathbf{0}) = \mathbf{b}$.

The transformation is linear exactly when $\mathbf{b} = \mathbf{0}$.

A nonzero translation fails additivity

Let $T(\mathbf{x}) = \mathbf{x} + \mathbf{b}$, where $\mathbf{b} \neq \mathbf{0}$. For arbitrary $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, additivity fails because

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= \mathbf{u} + \mathbf{v} + \mathbf{b}, \\ T(\mathbf{u}) + T(\mathbf{v}) &= (\mathbf{u} + \mathbf{b}) + (\mathbf{v} + \mathbf{b}) = \mathbf{u} + \mathbf{v} + 2\mathbf{b}. \end{aligned}$$

Therefore

$$T(\mathbf{u} + \mathbf{v}) \neq T(\mathbf{u}) + T(\mathbf{v}) \quad \text{when } \mathbf{b} \neq \mathbf{0}.$$

A nonzero translation fails scalar multiplication

Let $T(\mathbf{x}) = \mathbf{x} + \mathbf{b}$, where $\mathbf{b} \neq \mathbf{0}$.

Scalar multiplication also fails. For $\alpha \in \mathbb{R}$,

$$T(\alpha \mathbf{u}) = \alpha \mathbf{u} + \mathbf{b}, \quad \alpha T(\mathbf{u}) = \alpha \mathbf{u} + \alpha \mathbf{b}.$$

Hence

$$T(\alpha \mathbf{u}) \neq \alpha T(\mathbf{u}) \quad \text{when } \mathbf{b} \neq \mathbf{0} \text{ and } \alpha \neq 1.$$

In particular, $\alpha = 0$ gives $T(\mathbf{0}) = \mathbf{b} \neq \mathbf{0}$.

Homogeneous translation satisfies additivity

Represent a two-dimensional point in homogeneous coordinates, then extend translation by $\mathbf{b} = (b_1, b_2)^\top$ to all of $\mathbb{R}^3$:

$$\begin{aligned}\mathbf{x} &= (x_1, x_2)^\top \longmapsto \mathbf{x}_h = (x_1, x_2, 1)^\top, \\ T_h((x_1, x_2, w)^\top) &= (x_1 + wb_1, x_2 + wb_2, w)^\top.\end{aligned}$$

Let $\mathbf{u} = (u_1, u_2, s)^\top$ and $\mathbf{v} = (v_1, v_2, t)^\top$. Then

$$\begin{aligned}T_h(\mathbf{u} + \mathbf{v}) &= T_h((u_1 + v_1, u_2 + v_2, s + t)^\top) \\ &= (u_1 + v_1 + (s + t)b_1, u_2 + v_2 + (s + t)b_2, s + t)^\top,\end{aligned}$$

whereas

$$\begin{aligned}T_h(\mathbf{u}) + T_h(\mathbf{v}) &= (u_1 + sb_1, u_2 + sb_2, s)^\top \\ &\quad + (v_1 + tb_1, v_2 + tb_2, t)^\top \\ &= (u_1 + v_1 + (s + t)b_1, u_2 + v_2 + (s + t)b_2, s + t)^\top.\end{aligned}$$

Therefore $T_h(\mathbf{u} + \mathbf{v}) = T_h(\mathbf{u}) + T_h(\mathbf{v})$.

Homogeneous translation satisfies scalar multiplication

For convenience, restate the homogeneous-coordinate mapping:

$$T_h((x_1, x_2, w)^T) = (x_1 + wb_1, x_2 + wb_2, w)^T.$$

Let $\mathbf{u} = (u_1, u_2, s)^T \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$. Then

$$\begin{aligned} T_h(\alpha \mathbf{u}) &= T_h((\alpha u_1, \alpha u_2, \alpha s)^T) \\ &= (\alpha u_1 + \alpha s b_1, \alpha u_2 + \alpha s b_2, \alpha s)^T, \end{aligned}$$

whereas

$$\begin{aligned} \alpha T_h(\mathbf{u}) &= \alpha(u_1 + s b_1, u_2 + s b_2, s)^T \\ &= (\alpha u_1 + \alpha s b_1, \alpha u_2 + \alpha s b_2, \alpha s)^T. \end{aligned}$$

Therefore $T_h(\alpha \mathbf{u}) = \alpha T_h(\mathbf{u})$.

Normalised points use the plane $w = 1$

Represent a point $\mathbf{x} \in \mathbb{R}^n$ by

$$\mathbf{x}_h = \begin{pmatrix} \mathbf{x} \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}.$$

For the general affine transformation $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$,

$$T_h \begin{pmatrix} \mathbf{x} \\ w \end{pmatrix} = \begin{pmatrix} A & \mathbf{b} \\ \mathbf{0}^T & 1 \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ w \end{pmatrix} = \begin{pmatrix} A\mathbf{x} + w\mathbf{b} \\ w \end{pmatrix}.$$

Setting $w = 1$ recovers the original affine transformation:

$$T_h(\mathbf{x}_h) = \begin{pmatrix} A\mathbf{x} + \mathbf{b} \\ 1 \end{pmatrix}.$$

One distinction

Normalised points have $w = 1$. Their sum has $w = 2$, so the linearity rules apply to all of $\mathbb{R}^{n+1}$, not only to the plane of normalised points.

Scaling of the unit square

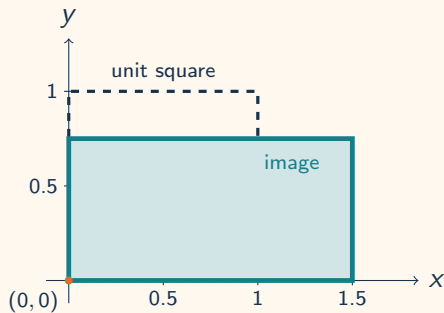
Scale horizontally by $\frac{3}{2}$ and vertically by $\frac{3}{4}$:

$$S = \begin{pmatrix} \frac{3}{2} & 0 & 0 \\ 0 & \frac{3}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus

$$(x, y) \mapsto \left(\frac{3}{2}x, \frac{3}{4}y\right).$$

The origin remains fixed. The unit square becomes a rectangle.

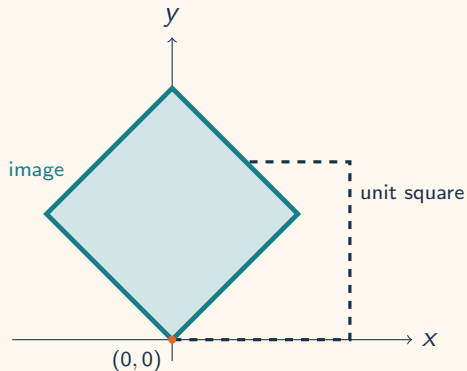


Rotation of the unit square

Rotate counterclockwise through 45° about the origin:

$$R = \begin{pmatrix} \cos(45^\circ) & -\sin(45^\circ) & 0 \\ \sin(45^\circ) & \cos(45^\circ) & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The origin remains fixed. Lengths and angles do not change.



Horizontal shear of the unit square

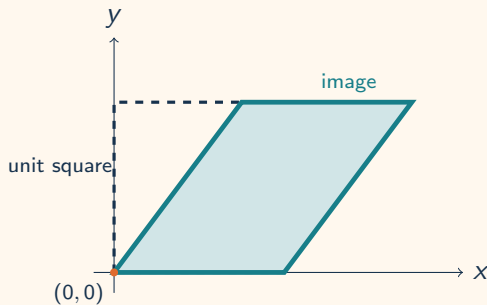
Apply a horizontal shear with factor $h = \frac{3}{4}$:

$$H = \begin{pmatrix} 1 & \frac{3}{4} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus

$$(x, y) \mapsto \left(x + \frac{3}{4}y, y\right).$$

The base stays fixed while the top edge moves to the right.



Translation of the unit square

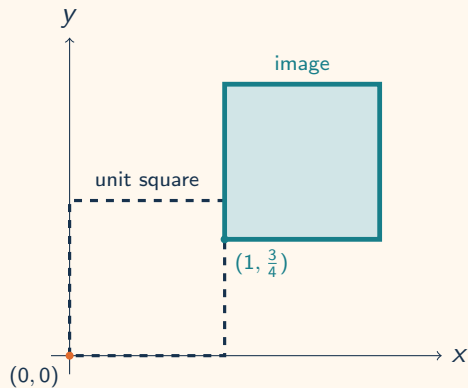
Translate by $(1, \frac{3}{4})$:

$$T = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & \frac{3}{4} \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus

$$(x, y) \mapsto (x + 1, y + \frac{3}{4}).$$

The square keeps its size and orientation, but its bottom-left corner moves from $(0, 0)$ to $(1, \frac{3}{4})$.



Matrix order follows the order of composition

Suppose an object is scaled, then sheared, and finally rotated:

$$\mathbf{x} \xrightarrow{S} S\mathbf{x} \xrightarrow{H} HS\mathbf{x} \xrightarrow{R} RHS\mathbf{x}.$$

The composite matrix is

$$M = RHS.$$

Read the product from right to left

The matrix nearest the vector acts first. Matrix multiplication is generally not commutative, so changing the order can change the result.

With homogeneous matrices, translations enter the same product. For example, rotation followed by translation has matrix $M = T_{\mathbf{b}}R_{\theta}$.

Example: rotate a square and then translate it

Rotate by 45° about the origin, then translate by $(2, 3)$:

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 2 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

For the vertex $(1, 1)$,

$$M \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 + \sqrt{2} \\ 1 \end{pmatrix}.$$

The rotation changes the square's orientation. The translation then moves its centre from $(0, 0)$ to $(2, 3)$.

Rotation about an arbitrary point uses three steps

Let $\mathbf{p} = (x_0, y_0)^T$. To rotate through θ about $\mathbf{p}$:

translate by $-\mathbf{p}$, rotate about the origin, translate by $\mathbf{p}$.

Thus

$$T(\mathbf{x}) = R_\theta(\mathbf{x} - \mathbf{p}) + \mathbf{p} = R_\theta\mathbf{x} + (I - R_\theta)\mathbf{p}.$$

In homogeneous coordinates, the composite matrix is

$$M = T_{\mathbf{p}} R_\theta T_{-\mathbf{p}}.$$

Check the centre

$T(\mathbf{p}) = R_\theta(\mathbf{p} - \mathbf{p}) + \mathbf{p} = \mathbf{p}$, so the centre of rotation remains fixed.

Images of the origin and basis vectors determine the map

Suppose an affine map sends

$$(0, 0) \mapsto (4, 5), \quad (1, 0) \mapsto (3, 0), \quad (0, 1) \mapsto (-1, 2).$$

For $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, the image of the origin gives

$$\mathbf{b} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}.$$

Subtract $\mathbf{b}$ from the other two images to obtain the columns of A :

$$A\mathbf{e}_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix}, \quad A\mathbf{e}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 4 \\ 5 \end{pmatrix} = \begin{pmatrix} -5 \\ -3 \end{pmatrix}.$$

Therefore

$$A = \begin{pmatrix} -1 & -5 \\ -5 & -3 \end{pmatrix}, \quad T(x, y) = (-x - 5y + 4, -5x - 3y + 5).$$

The same homogeneous block form works in three dimensions

For $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ on $\mathbb{R}^3$, use

$$M = \begin{pmatrix} A & \mathbf{b} \\ \mathbf{0}^\top & 1 \end{pmatrix}.$$

For example, translation by (b_x, b_y, b_z) has matrix

$$\begin{pmatrix} 1 & 0 & 0 & b_x \\ 0 & 1 & 0 & b_y \\ 0 & 0 & 1 & b_z \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Scaling, shearing and rotation use the upper-left 3×3 block. The last column remains the translation vector.

Composition still uses multiplication

Translation by $(0, 2, 5)$ followed by translation by $(1, 4, 3)$ gives translation by $(1, 6, 8)$.

Short Revision Quiz

Affine transformations

Question 1 — Affine form

Which expression describes a general affine transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$?

- A. $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$
- B. $T(\mathbf{x}) = A + \mathbf{x}\mathbf{b}^T$
- C. $T(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$

Question 2 — Translation and linearity

Let $T(\mathbf{x}) = \mathbf{x} + \mathbf{b}$, where $\mathbf{b} \neq \mathbf{0}$. Which statement proves that T is not linear?

- A. $T(\mathbf{0}) = \mathbf{0}$
- B. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
- C. $T(\mathbf{0}) = \mathbf{b} \neq \mathbf{0}$

Question 3 — Homogeneous translation

Which matrix translates points in $\mathbb{R}^2$ by (p, q) ?

A. $\begin{pmatrix} 1 & p & 0 \\ q & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

B. $\begin{pmatrix} 1 & 0 & p \\ 0 & 1 & q \\ 0 & 0 & 1 \end{pmatrix}$

C. $\begin{pmatrix} p & 0 & 1 \\ 0 & q & 1 \\ 0 & 0 & 1 \end{pmatrix}$

Question 4 — Homogeneous linearity

Why does the homogeneous-coordinate transformation $T_h(\mathbf{z}) = M\mathbf{z}$ satisfy the linearity rules?

- A. Every homogeneous vector has last coordinate 1
- B. Multiplication by a fixed matrix is a linear transformation on $\mathbb{R}^{n+1}$
- C. The translation vector becomes zero

Question 5 — Order of composition

An object is first scaled by S , then rotated by R , and finally translated by $T_{\mathbf{b}}$. Which matrix represents the composition?

A. $ST_{\mathbf{b}}R$

B. $T_{\mathbf{b}}RS$

C. $SRT_{\mathbf{b}}$

Question 6 — Rotation about a point

Which homogeneous product rotates about the point $\mathbf{p}$ rather than the origin?

A. $T_{\mathbf{p}} R_{\theta} T_{-\mathbf{p}}$

B. $T_{-\mathbf{p}} R_{\theta} T_{\mathbf{p}}$

C. $R_{\theta} T_{\mathbf{p}} T_{-\mathbf{p}}$

Question 7 — Recovering an affine map

If $T(\mathbf{0}) = \mathbf{b}$, what is the j th column of A in $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$?

- A. $T(\mathbf{e}_j)$
- B. $T(\mathbf{e}_j) - \mathbf{b}$
- C. $T(\mathbf{e}_j) + \mathbf{b}$

Question 8 — Two translations

Translation by $\mathbf{b}_2$ is followed by translation by $\mathbf{b}_1$. What is the resulting translation vector?

A. $\mathbf{b}_1 - \mathbf{b}_2$

B. $\mathbf{b}_1 + \mathbf{b}_2$

C. $\mathbf{b}_1^T \mathbf{b}_2$

Quiz answers 1–4

1 — A

An affine map has a linear part $A\mathbf{x}$ and a translation $\mathbf{b}$.

2 — C

Every linear transformation maps the zero vector to itself. A nonzero translation maps $\mathbf{0}$ to $\mathbf{b}$.

3 — B

The translation entries occupy the last column of the homogeneous matrix.

4 — B

T_h multiplies arbitrary vectors in $\mathbb{R}^{n+1}$ by a fixed matrix, so it preserves addition and scalar multiplication.

Quiz answers 5–7

5 — B

The rightmost matrix acts first, so scaling, rotation and translation give $T_{\mathbf{b}}RS$.

6 — A

Move $\mathbf{p}$ to the origin, rotate, then move it back: $T_{\mathbf{p}}R_{\theta}T_{-\mathbf{p}}$.

7 — B

$T(\mathbf{e}_j) = A\mathbf{e}_j + \mathbf{b}$, so subtracting $\mathbf{b}$ gives the j th column of A .

Quiz answer 8

8 — B

Translations add: $(\mathbf{x} + \mathbf{b}_2) + \mathbf{b}_1 = \mathbf{x} + (\mathbf{b}_1 + \mathbf{b}_2)$.

Final takeaway

Homogeneous coordinates turn every affine transformation into one matrix transformation, which makes composition systematic while preserving the distinction between points and vectors.